COURSE SYLLABUS

WEEK	SUBJECT	DATE	
1	INTRODUCTION BASIC DEFINITIONS (TERMS)	05.10.2020 08.10.2020	Monday Thursday
2	ROAD USER CHARECTERISTICS VEHICLE MOVEMENTS	12.10.2020 15.10.2020	Monday Thursday
3	TRAFFIC FLOW AND CAPACITY ROADWAY CLASSIFICATION AND GEOMETRIC STANDARDS	19.10.2020 22.10.2020	Monday Thursday
4	HIGHWAY ROUTE STUDY **NATIONAL HOLIDAY**	26.10.2020 29.10.2020	Monday Thursday
5	HORIZANTAL CURVE AND SUPERELEVATION (Project sheet distribution) HORIZANTAL CURVE AND SUPERELEVATION	02.11.2020 05.11.2020	Monday Thursday
6	APPLICATION (VEHICLE MOVEMENTS) ASSIGNMENT - APPLICATION (TRAFFIC)	09.11.2020 12.11.2020	Monday Thursday
7	APPLICATION (HORIZANTAL CURVE) ASSIGNMENT-APPLICATION (HORIZANTAL CURVE)	16.11.2020 19.11.2020	Monday Thursday
8	**MID TERM EXAM**	23.11.2020 26.11.2020	Monday Thursday
9	TRANSITION CURVES- ASSIGNMENT (HAND IN) TRANSITION CURVES – ASSIGNMENT (RETURN)	30.11.2020 03.12.2020	Monday Thursday
10	VERTICAL ALIGNMENT AND VERTICAL CURVE VERTICAL ALIGNMENT AND VERTICAL CURVE	07.12.2020 10.12.2020	Monday Thursday
11	APPLICATION (TRANSITION CURVES) ASSIGNMENT	14.12.2020 17.12.2020	Monday Thursday
12	APPLICATION (VERTICAL CURVE) APPLICATION (VERTICAL AND HORIZANTAL CURVE)	21.12.2020 24.12.2020	Monday Thursday
13	PLANNING OF INTERSECTION AND URBAN ROADS PROJECTS DELIVERY DUE DATE (December 30, 2020)	28.12.2020 31.12.2020	Monday Thursday
14	DRAINAGE IN HIGHWAY **MAKEUP EXAM**	04.01.2020 07.01.2020	Monday Thursday

TRAFFIC FLOW, CAPACITY AND PROPERTIES OF THE TRAFFIC

Main variables of the traffic:

These variables are acquired by various methods but they are not used independently. The other variables can be acquired by using the main variables length **(X)** (m, km), time **(T)** (sec, min, h) and traffic unit **(N)** (veh).

TRAFFIC FLOW, CAPACITY AND PROPERTIES OF THE TRAFFIC

Fundemental Parameters of Traffic Flow;

Speed (km/h)

$$V = \frac{X}{T}$$





Volume (veh/h)

$$Q = \frac{N}{T}$$







Density (veh/km)

$$k = \frac{N}{X}$$



TRAFFIC FLOW, CAPACITY AND PROPERTIES OF THE TRAFFIC

Main variables of the traffic:

These variables are acquired by various methods but they are not used independently. The other variables can be acquired by using the main variables length **(X)** (m, km), time **(T)** (sec, min, h) and traffic unit **(N)** (veh).

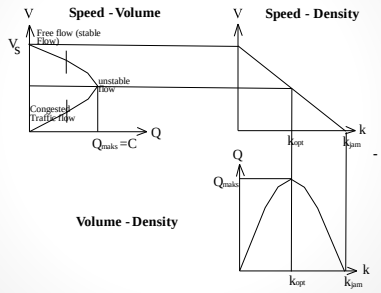
Derived variables of the traffic:

They are the variables derived from the main variables of the traffic to explain the traffic flow. They can be listed as

the volume **(q)** (veh/sec, veh/h),
the density **(k)** (veh/km) and
the speed **(u)** (m/sec, km/h).

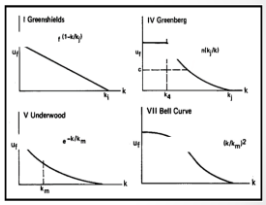
TRAFFIC FLOW, CAPACITY AND PROPERTIES OF THE TRAFFIC

$$\mathbf{Q = \frac{N}{L} \times \frac{L}{T}} \longrightarrow \mathbf{Q = k \times V}$$

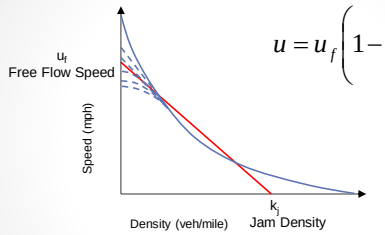


TRAFFIC FLOW, CAPACITY AND PROPERTIES OF THE TRAFFIC

	Speed-Volume Releatiınship
Greenshields	$u = u_f * \left(1 - \frac{k}{k_j}\right)$
Greenberg	$u = u_m * ln\left(\frac{k_j}{k}\right)$
Underwood	$u = u_f * e^{\frac{-k}{k_m}}$
Bell Curve	$u = u_f * e^{-\frac{1}{2}\left(\frac{k}{k_m}\right)^2}$



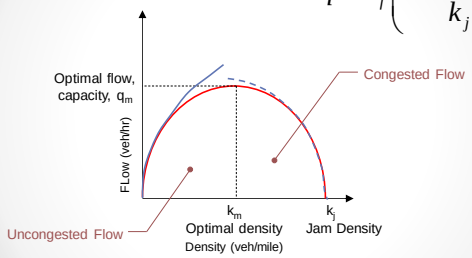
Speed - Density



$$u = u_f\left(1 - \frac{k}{k_j}\right)$$

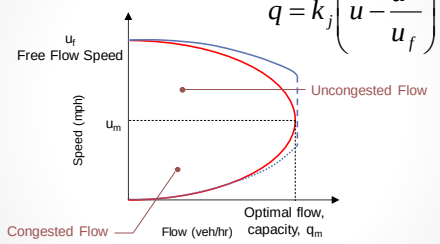
Volume - Density

$$q = u_f\left(k - \frac{k^2}{k_j}\right)$$



Speed - Volume

$$q = k_j\left(u - \frac{u^2}{u_f}\right)$$



TRAFFIC FLOW, CAPACITY AND PROPERTIES OF THE TRAFFIC

Peak hour factor (PHF) : The ratio between the measured traffic flow and the calculated traffic flow is named as the peak hour factor (Zirve Saat Faktörü-ZSF)

$q = \frac{N_t}{t} * 60$

N_t: number of veh. passing in a period of time t
t : time period(minutes)
q: Volume(veh/hour)

$$PHF = \frac{Q}{q_m}$$

Q: The number of veh. Passed in a whole time period

Time Period	Volume (vehicles)	Rate of Flow (vph)
4:00-4:15	1,200	4,800
4:15-4:30	1,400	5,600
4:30-4:45	1,700	6,800
4:45-5:00	1,500	6,000
4:00-5:00	5,800	

$$PHF = \frac{5800}{6800} = 0.853$$

TRAFFIC FLOW, CAPACITY AND PROPERTIES OF THE TRAFFIC

Speed, Flow, Capacity and Density

Speed (V): the distance travelled per unit of time. $V = \frac{L}{T}$

Commercial Speed: waiting and delays are all included into the time when calculating the speed. It is used as a performance of the urban traffic.
Spot Speed: the instantaneous speed of a vehicle at a specified location. Spot speed can be used to design the geometry of road like horizontal and vertical curves, super elevation etc.
Design Speed: Design speed is a selected speed used to determine the various geometric features of the roadway.
Operating Speed: The highest overall speed at which a driver can travel on a given highway under favourable weather conditions and under prevailing traffic conditions without at any time exceeding the safe speed as determined by the design speed on a section-by-section basis. In current practice, the term operating speed refers to the speed at which drivers are observed operating their vehicles.

TRAFFIC FLOW, CAPACITY AND PROPERTIES OF THE TRAFFIC

A ⟷ B

L= 5 km

V_{BA}= 100 km/sa
V_{AB}= 50 km/sa

V_{avg}= ?

$V_{avg} = \frac{100+50}{2}$ = 75 km/sa

t_{AB}= 0.1 sa = 6 dak.
t_{BA}= 0.05 sa = 3 dak.

$V_{ort} = \frac{5+5}{0.15}$ = 66.67 km/sa

TRAFFIC FLOW, CAPACITY AND PROPERTIES OF THE TRAFFIC

Time-mean speed U_t

Average value of speeds of vehicles passing a spot during some period of time (km/h)

$$v_t = \frac{1}{n}\sum_{i=1}^{n} v_i \qquad v_t = \frac{\sum_{i=1}^{n} q_i v_i}{\sum_{i=1}^{n} q_i}$$

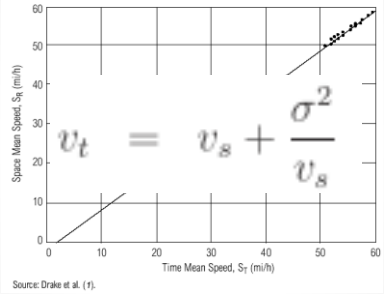
Space-mean speed U_s

Average value of speeds of vehicles along a road at some instant (km/h)

$$t_s = \frac{\Sigma t_i}{n} = \frac{1}{n}\Sigma\frac{1}{v_i} \qquad v_s = \frac{n}{\sum_{i=1}^{n}\frac{1}{v_i}} \qquad v_s = \frac{\sum_{i=1}^{n} q_i}{\sum_{i=1}^{n}\frac{q_i}{v_i}}$$

TRAFFIC FLOW, CAPACITY AND PROPERTIES OF THE TRAFFIC

EXHIBIT 7-1. TYPICAL RELATIONSHIP BETWEEN TIME MEAN AND SPACE MEAN SPEED



Source: Drake et al. (1).

TRAFFIC FLOW, CAPACITY AND PROPERTIES OF THE TRAFFIC

Flow rate (Q): The number of vehicles that pass from a highway section in a period of time is the flow rate. The period of time can be taken as the intervals like 5, 10 ,15 minutes. If the time interval is equal to 1 hour, It is called **traffic volume**.

$$Q = \frac{N}{T}$$

The capacity of a highway can be stated as the maximum number of vehicles that can pass on a single or double direction at a specific time period, from a section with specific properties under specific conditions.

Theoretical capacity: The maximum number of vehicles that can pass from a highway section within an hour on single or double direction under ideal road, traffic and control conditions is named as the theoretical capacity.

Practical capacity: The capacity value on dominant road, traffic and control conditions, without congestions or accidents on the traffic flow; the drivers have no restraints on speed selection, vehicle following distance determination or no restrictions on vehicle overtaking possibilities is the practical capacity.

TRAFFIC FLOW, CAPACITY AND PROPERTIES OF THE TRAFFIC

The affects that reduce the capacity:

i) Highway geometry related effects
- **a)** Insufficient lane width
- **b)** Insufficient shoulder width
- **c)** Insufficient auxiliary lanes
- **d)** Horizantal curve radius and grade
- **e)** Road surface condition.

ii) Traffic flow releated effects
- **a)** Vehicle composition
- **b)** Interruption on traffic
- **c)** Driver behaviour

iii) Other effects
- **a)** Sever weather condidition
- **b)** Traffic incidents

TRAFFIC FLOW, CAPACITY AND PROPERTIES OF THE TRAFFIC

Factors that influence traveller perceived quality of service have been found to include (HCM, 2010)

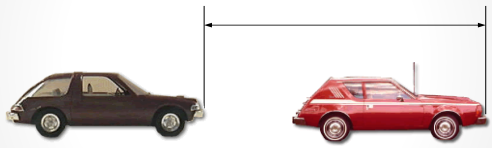
- Travel time, speed, and delay;
- Number of stops incurred;
- Travel time reliability;
- Manoeuvrability (e.g., ease of lane changing, percent time-spent-following other vehicles);
- Comfort (e.g., bicycle and pedestrian interaction with and separation from traffic, transit vehicle crowding, ride comfort);
- Convenience (e.g., directness of route, frequency of transit service);
- Safety (actual or perceived);
- User cost;
- Availability of facilities and services;
- Facility aesthetics; and
- Information availability (e.g., highway wayfinding signage, transit route and schedule information).

Traffic Flow

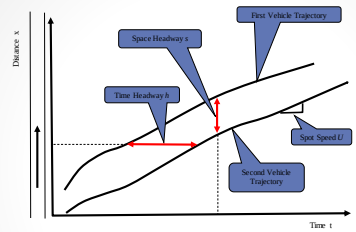
1. Basic Concepts
   a. Traffic Flow
   b. Headway
   c. Speed
   d. Density
2. Fundemental Relations of Traffic Flow

Traffic Flow

Distance between consecutive vehicles



Traffic Flow



Time headway h = time measured at a spot between two consecutive vehicle passages

Space headway s = distance between two consecutive vehicles at an instant

Speed U = instantaneous speed of a vehicle at a spot and at an instant

Traffic Flow

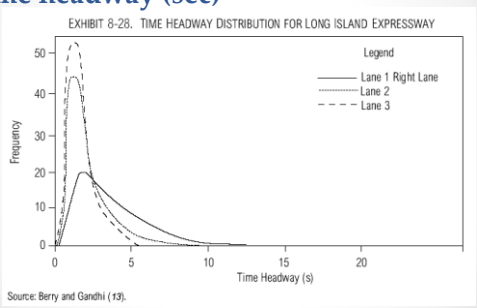
Time headway (sec)

- h; time measured at a spot between two consecutive vehicle passages (sec.)

$$t = \sum_{i=1}^{n} h_i \qquad q = \frac{n}{\sum_{i=1}^{n} h_i} = \frac{1}{\overline{h}}$$

Traffic Flow

Time headway (sec)

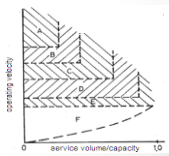


TRAFFIC FLOW, CAPACITY AND PROPERTIES OF THE TRAFFIC

Level of Service (LOS): The level of service, in general is the condition of the given lane or platform under various traffic volumes. And, it is developed in US for HCM. It is defined, «LOS is a quantitative stratification of a performance measure or measures that represent quality of service. The measures used to determine LOS for transportation system elements are called service measures» in HCM2010.

The HCM defines six levels of service, ranging from A to F, for each service measure, or for the output from a mathematical model based on multiple performance measures. LOS A represents the best operating conditions from the traveller's perspective and LOS F the worst.

Level of service	General operating conditions
A	Free flow
B	Reasonably free flow
C	Stable flow
D	Approaching unstable flow
E	Unstable flow
F	Forced or breakdown flow



TRAFFIC FLOW, CAPACITY AND PROPERTIES OF THE TRAFFIC

Level of service A: Shows the free flow conditions on which the drivers are not affected by each other during drive. Drivers are free to select their velocities and making overtake and lane change manouevres unless there's a speed or sight distance constraint. Travel is very comfortable. Average speed is around 100 km/h. The speed on level of service A is named as the **free speed**. Volume/capacity ratio is lower than 15%. Service volume is below 420 veh/h as the sum of the two directions. Delay percentage is below 30%.



TRAFFIC FLOW, CAPACITY AND PROPERTIES OF THE TRAFFIC

Level of service B: Traffic flow is steady on this level. There is not an important constraint on the speed selection and manouevre freedom of the drivers. An important speed decrease is not montiored. Travel comfort is good although not as good as the free flow. Average speed is around 90 km/h, average delay is below 45%, volume/capacity ratio is below 25%, the service volume for the two directions is below 750 veh/h.



TRAFFIC FLOW, CAPACITY AND PROPERTIES OF THE TRAFFIC

Level of service C: Although the steady condition of the traffic flow continues; the effect of the low-speed vehicles starts to be felt. So the speed selection and the manouevre freedom of the drivers starts to be constrained. Travel comfort and speed starts to decrease. This condition shows that the traffic congestions may occur. Average speed is around 85 km/h, the maximum value of the delay is 60%, volume/capacity ratio is around 40% or less, the service volume for two directions is around 1.200 veh/h or less.



TRAFFIC FLOW, CAPACITY AND PROPERTIES OF THE TRAFFIC

Level of service D: The traffic flow is closer to the unsteady state. The speed selection and the manouevre freedom of the drivers are constrained as an effect of other drivers. Travel comfort and velocities continue to decrease. The effect of the overtaking, merging and diverging vehicles is high. Also congestions and queings start to be seen. The limit of the average speed is 80 km/h, delay is around 75%. Volume/capacity ratio is around 65% and the service volume for two directions is around 1.800 veh/h or less. This level of service is acceptable for suburban highways.



TRAFFIC FLOW, CAPACITY AND PROPERTIES OF THE TRAFFIC

Level of service E: Traffic flow is unsteady. Volume/capacity ratio is very close to 100%. The drivers' ability to select their speed or to manouevre is totally constrained. Traffic flow is almost congested and the manouevre of the vehicles feed this congestion. Average speed is below 80 km/h and on long ascending sections below 40 km/h. Delay is over 75%. The service volume for two directions is around 2.800 veh/h and very close to the capacity of the road.



TRAFFIC FLOW, CAPACITY AND PROPERTIES OF THE TRAFFIC

Level of service F: Volume/capacity ratio is over 100%. In other words, the demand for the road is higher than its capacity. Delay is very close to 100%. As the traffic is totally congested, it tries to flow forced. Level of service F shows the limit case on which the road can no longer serve.



TRAFFIC FLOW, CAPACITY AND PROPERTIES OF THE TRAFFIC

LOS A



LOS B



Los C



LOS D



LOS E

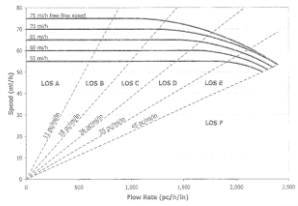


LOS F



TRAFFIC FLOW, CAPACITY AND PROPERTIES OF THE TRAFFIC

Exhibit 11-6
LOS for Basic Freeway Segments



Equation 10-1

$$FFS = 75.4 - f_{LW} - f_{LC} - 3.22\,TRD^{0.84}$$

where

FFS = free-flow speed (mi/h),
f_{LW} = adjustment for lane width (mi/h),
f_{LC} = adjustment for lateral clearance (mi/h), and
TRD = total ramp density (ramps/mi).

TRAFFIC FLOW, CAPACITY AND PROPERTIES OF THE TRAFFIC

Level of Service	Density (pc/mi/ln)
A	≤11
B	>11–18
C	>18–26
D	>26–35
E	>35–45
F	>45 or any component v_p/C ratio > 1.00

Exhibit 10-7
LOS Criteria for Freeway Facilities

TRAFFIC FLOW, CAPACITY AND PROPERTIES OF THE TRAFFIC

EXHIBIT 16-2. LOS CRITERIA FOR SIGNALIZED INTERSECTIONS

LOS	Control Delay per Vehicle (s/veh)
A	≤ 10
B	> 10–20
C	> 20–35
D	> 35–55
E	> 55–80
F	> 80

TRAFFIC FLOW, CAPACITY AND PROPERTIES OF THE TRAFFIC

EXHIBIT 15-2. URBAN STREET LOS BY CLASS

Urban Street Class	I	II	III	IV
Range of free-flow speeds (FFS)	90 to 70 km/h	70 to 55 km/h	55 to 50 km/h	55 to 40 km/h
Typical FFS	80 km/h	65 km/h	55 km/h	45 km/h
LOS	Average Travel Speed (km/h)			
A	> 72	> 59	> 50	> 41
B	> 56–72	> 46–59	> 39–50	> 32–41
C	> 40–56	> 33–46	> 28–39	> 23–32
D	> 32–40	> 26–33	> 22–28	> 18–23
E	> 26–32	> 21–26	> 17–22	> 14–18
F	≤ 26	≤ 21	≤ 17	≤ 14

34

TRAFFIC FLOW, CAPACITY AND PROPERTIES OF THE TRAFFIC

Severity of terrain can affect the operating capabilities of individual vehicles in the traffic stream

Level Terrain: Any combination of grades and horizontal or vertical alignment that permits heavy vehicles to maintain the same speed as passenger cars, typically containing short grades of no more than 2%.

Rolling terrain: Any combination of grades and horizontal or vertical alignment that causes heavy vehicles to reduce their speed substantially below that of passenger cars but that does not cause heavy vehicles to operate at crawl speeds for any significant length of time or at frequent intervals.

Mountainous terrain: Any combination of grades and horizontal and vertical alignment that causes heavy vehicles to operate at crawl speed for significant distances or at frequent intervals.

TRAFFIC FLOW, CAPACITY AND PROPERTIES OF THE TRAFFIC

Level of Service Criteria on two lane two direction suburban highways

Level of Service	Data Percentage (%)	Average Velocity (km/h	Volume/Capacity Ratio					
			The short section percentage on which the sight distance is below 450m					
Flat			0	20	40	60	80	100
A	< 30	> 93	0,15	0,12	0,09	0,07	0,05	0,04
B	< 45	> 88	0,27	0,24	0,21	0,19	0,17	0,16
C	< 60	> 83	0,43	0,39	0,36	0,34	0,33	0,32
D	< 75	> 80	0,64	0,62	0,60	0,59	0,58	0,57
E	> 75	> 72	1,00	1,00	1,00	1,00	1,00	1,00
F	100	< 72	-	-	-	-	-	-
Undulating								
A	< 30	> 91	0,15	0,10	0,07	0,05	0,04	0,03
B	< 45	> 86	0,26	0,23	0,19	0,17	0,15	0,13
C	< 60	> 82	0,42	0,39	0,35	0,32	0,30	0,28
D	< 75	> 78	0,62	0,57	0,52	0,48	0,46	0,43
E	> 75	> 64	0,97	0,94	0,92	0,91	0,90	0,90
F	100	< 64	-	-	-	-	-	-
Mountainous								
A	< 30	> 90	0,14	0,09	0,07	0,04	0,02	0,01
B	< 45	> 86	0,25	0,20	0,16	0,13	0,12	0,10
C	< 60	> 78	0,39	0,33	0,28	0,23	0,20	0,16
D	< 75	> 72	0,58	0,50	0,45	0,40	0,37	0,33
E	> 75	> 56	0,91	0,87	0,84	0,82	0,80	0,78
F	100	< 56	-	-	-	-	-	-

TRAFFIC FLOW, CAPACITY AND PROPERTIES OF THE TRAFFIC

Traffic Flow Rate;

The level of service on two lane two direction highways

$$SF_i = 2.800 * \left(\frac{Q}{C}\right)_i * f_d * f_w * f_{HV}$$

Sf_i: Two direction traffic flow rate for i[th] LOS (veh./h)

$(Q/C)_i$:The volume/capacity ratio for level of service (i)

f_d : The direction distribution factor of the traffic

f_w : The adjustment factor about the lane and banquet width

f_{HV}: The adjustment factor about the heavy vehicle traffic

$$fHV = \frac{1}{\left[1 + P_T * (E_T - 1) + P_B * (E_B - 1)\right]}$$

P_T, P_B : The percentage of trucks AND BUSES on the

E_T, E_B : Unit vehicle equivalent of the truck and bus

TRAFFIC FLOW, CAPACITY AND PROPERTIES OF THE TRAFFIC

The direction distribution factor of the traffic

Directions distribution fact.	100/0	90/10	80/20	70/30	60/40	50/50
f_d	0,71	0,75	0,83	0,89	0,94	1,00

The adjustment factor about the lane and banquet width

Lane Width (m)	3,7		3,3		3,0		2,7	
Banquet Width (m)	LOS							
	A-D	E	A-D	E	A-D	E	A-D	E
>=2	1,00	1,00	0,93	0,94	0,84	0,87	0,70	0,76
1	0,89	0,96	0,82	0,91	0,75	0,84	0,63	0,73
0	0,70	0,88	0,65	0,82	0,58	0,75	0,49	0,66

TRAFFIC FLOW, CAPACITY AND PROPERTIES OF THE TRAFFIC

The service volume on multi lane highways;

$$SF_i = C_j * \left(\frac{Q}{C}\right)_i * N * f_p * f_w * f_{HV} * f_e$$

SF_i : The service volume of a single direction for level of service (i) under dominant road and traffic conditions (veh/h)

Cj: The capacity per lane under ideal conditions for project speed (j).

Vp=100-110 km/h 2.000 veh./h

Vp=80 km/h 1.900 heh./h

$(Q/C)_i$: The volume/capacity rate fot the level of service (i)

f_p :The adjustment factor about the knowledge of the highway for the driver.

f_e : The adjustment factor about the environmental conditions of the highway route

f_w : The adjustment factor about the lane and banquet width

f_{HV} : The adjustment factor about the heavy vehicle traffic

TRAFFIC FLOW, CAPACITY AND PROPERTIES OF THE TRAFFIC

Level of service criteria on multi lane highways

Level of Service	Density (auto/km/ln)	Vp=110 km/h			Vp=100 km/h			Vp=80 km/h		
		$V_{ave.}$ (km/h)	Q/C	q_m (vh/h/ln)	$V_{ave.}$ (km/h)	Q/C	q_m (vh/h/ln)	$V_{ave.}$ (km/h)	Q/C	q_m (vh/h/ln)
A	≤ 7,5	≥ 91	0,36	700	≥ 80	0,33	650	-	-	-
B	≤ 12,5	≥ 85	0,54	1.100	≥ 77	0,50	1.000	≥ 67	0,45	850
C	≤ 18,8	≥ 80	0,71	1.400	≥ 70	0,65	1.300	≥ 62	0,60	1.150
D	≤ 26,3	≥ 64	0,87	1.750	≥ 64	0,80	1.600	≥ 56	0,76	1.450
E	≤ 41,9	≥ 48	1,00	2.000	≥ 48	1,00	2.000	≥ 45	1,00	1.900
F	> 41,9	< 48	v	v	< 48	v	v	< 45	v	v

v: Varies

TRAFFIC FLOW, CAPACITY AND PROPERTIES OF THE TRAFFIC

Service volume on multi lane highways

Level of Service	Unadjusted service volume (for a single lane (auto/h/ln) Truck Percentage(%)				
	0	5	10	15	20
Flat Terrain					
A	700	700	650	650	650
B	1.100	1.050	1.000	1.000	1.000
C	1.400	1.350	1.300	1.250	1.250
D	1.750	1.700	1.650	1.600	1.550
E	2.000	1.950	1.850	1.800	1.750
Undulating Terrain					
A	700	600	550	500	500
B	1.100	950	850	800	700
C	1.400	1.200	1.100	1.000	900
D	1.750	1.500	1.350	1.250	1.100
E	2.000	1.750	1.550	1.450	1.250
Mountainous Terrain					
A	700	500	400	350	300
B	1.100	800	650	550	450
C	1.400	1.050	850	700	600
D	1.750	1.300	1.050	850	750
E	2.000	1.500	1.200	1.000	850

TRAFFIC FLOW, CAPACITY AND PROPERTIES OF THE TRAFFIC

The adjustment factors about lanes and banquets for multi lane highways (f_w)

Distance to Obstacle (m)	Adjustment factors (f_w)							
	Obstacle on one side				Obstacle on both sides			
	Lane width (m)							
	3,7	3,3	3,0	2,7	3,7	3,3	3,0	2,7
Divided multi lane highway - Two lanes each side								
≥ 2,0	1,00	0,97	0,91	0,81	1,00	0,97	0,91	0,81
1,0	0,98	0,95	0,89	0,80	0,97	0,94	0,88	0,78
0	0,90	0,87	0,82	0,73	0,81	0,79	0,74	0,66
Divided multi lane highway - Three lanes each side								
≥ 2,0	1,00	0,96	0,89	0,78	1,00	0,96	0,89	0,78
1,0	0,98	0,94	0,88	0,77	0,97	0,93	0,86	0,76
0	0,94	0,91	0,85	0,74	0,91	0,87	0,81	0,70
Undivided multi lane highway - Two lanes each side								
≥ 2,0	1,00	0,95	0,89	0,77	N	N	N	N
1,0	0,97	0,93	0,87	0,76	N	N	N	N
0	0,88	0,85	0,80	0,70	0,81	0,79	0,74	0,66
Undivided multi lane highway - Three lanes each side								
≥ 2,0	1,00	0,95	0,89	0,77	N	N	N	N
1,0	0,98	0,94	0,88	0,76	N	N	N	N
0	0,94	0,90	0,83	0,72	0,91	0,87	0,81	0,70

* If the obstacle on each side is at different destination, the average is taken
N: The factors for the obstacle on one side is taken

TRAFFIC FLOW, CAPACITY AND PROPERTIES OF THE TRAFFIC

The adjustment factors the represen t the environmental properties (f_e)

Settlement type	Highway type	
	Divided	Undivided
Rural	1,00	0,95
Urban	0,90	0,80

Driver factors (f_p)

Driver type	(f_p)
Regular user	1,00
Occassional user	0,75-0,90

Unit vehicle equivalents of the trucks and busses

Vehicle type	Terrain type		
	Flat	Undulaced	Mountainous
Truck (E_T)	1,7	4,0	8,0
Bus (E_B)	1,5	3,0	5,0

TRAFFIC FLOW, CAPACITY AND PROPERTIES OF THE TRAFFIC

The adviced level of services at the end of service lifes for suburban highways

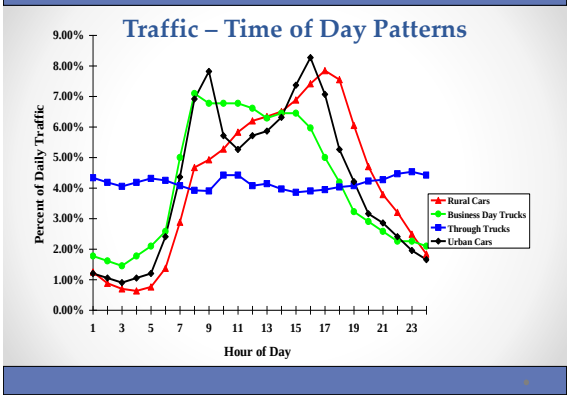
Highway types	Terrain type		
	Flat	Undulaced	Mountainous
Multi lane highways	B	B	B - C
Two lane highways			
1st class highways	B	B - C	C
2nd class highways	C	C - D	D
3rd class highways	C	D	D
4th class highways	D	D	D

TRAFFIC FLOW, CAPACITY AND PROPERTIES OF THE TRAFFIC

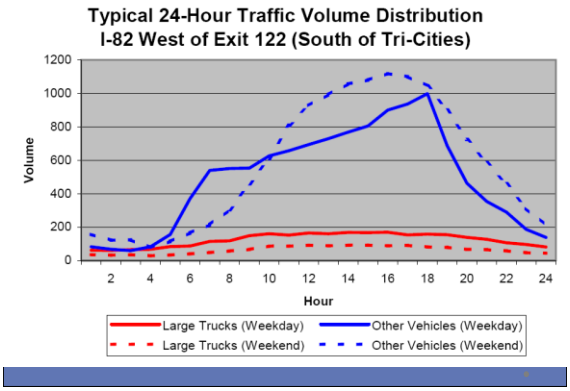
The adviced level of services at the end of the service lifes of urban highways

Highway types	Terrain type		
	Flat	Undulaced	Mountainous
Peripheral roads			
Multi lane highways	C	C	C
Two lane highways	C	C	C
Urban roads			
Multi lane highways	C	C	C
Two lane highways	C - D	C - D	C -D

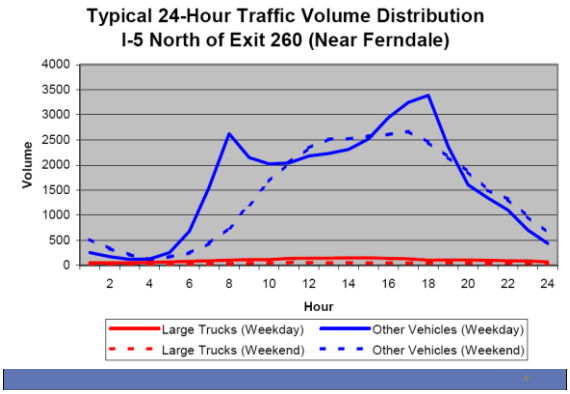
TRAFFIC FLOW, CAPACITY AND PROPERTIES OF THE TRAFFIC



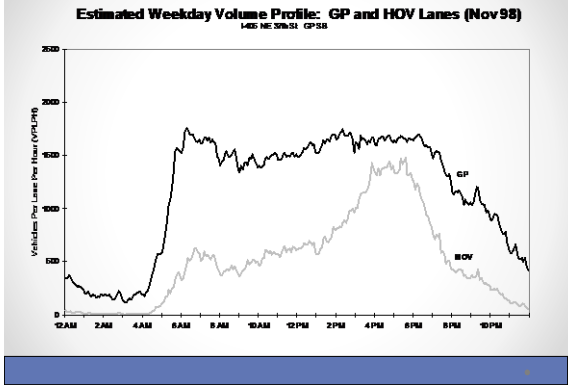
TRAFFIC FLOW, CAPACITY AND PROPERTIES OF THE TRAFFIC



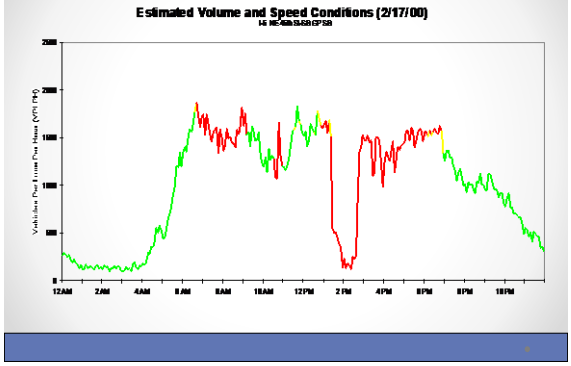
TRAFFIC FLOW, CAPACITY AND PROPERTIES OF THE TRAFFIC



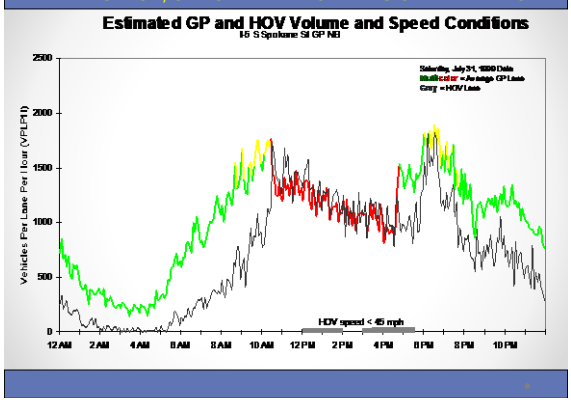
TRAFFIC FLOW, CAPACITY AND PROPERTIES OF THE TRAFFIC



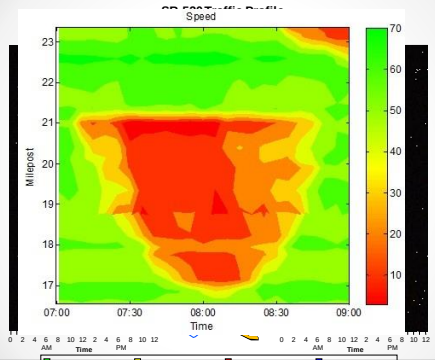
TRAFFIC FLOW, CAPACITY AND PROPERTIES OF THE TRAFFIC



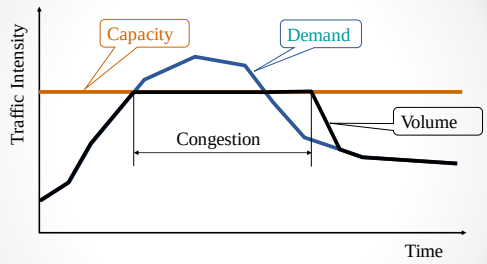
TRAFFIC FLOW, CAPACITY AND PROPERTIES OF THE TRAFFIC



TRAFFIC FLOW, CAPACITY AND PROPERTIES OF THE TRAFFIC



TRAFFIC FLOW, CAPACITY AND PROPERTIES OF THE TRAFFIC



TRAFFIC FLOW, CAPACITY AND PROPERTIES OF THE TRAFFIC

Some issues releated with capacity and LOS of a roadway;

i) What is a service life of a existing highway in a certain LOS? ($Q<C_i$)

ii) What type of improvement could be done, if traffic is higher than capacity? ($Q \geq C_i$)

iii) How many lanes should be added to improve the LOS of a highway?

iv) How many lanes are required to maintain the certain LOS in a highway for a project life?

TRAFFIC FLOW, CAPACITY AND PROPERTIES OF THE TRAFFIC

Project traffic: The traffic volume that is used to design a highway

Project life or the **service life**: The time period in which the highway is expected to provide utility

Annual average daily traffic (AADT): Average daily traffic is the average number of vehicles that pass from a highway section during a day. This is given by the total no. of vehicles passing through a section in a year divided by 365.

Directional distribution factor (D)

Directional Design hourly volume (DDHV)

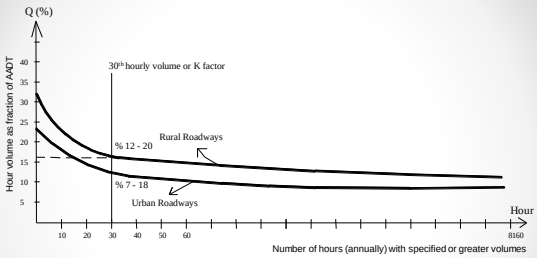
Design hourly volume (DHV): Traffic volume used for design calculations. Typically between the 10th and 50th highest volume hour of the year (30th highest is most common)

$$DDHV = K \times D \times AADT$$

TRAFFIC FLOW, CAPACITY AND PROPERTIES OF THE TRAFFIC

Number of hours (annually) with specified or greater volumes n	Vehicle/hour	Hour volume as fraction of AADT Q/AADT (%)
1	Q_1	
2	Q_2	
3	Q_3	
10	Q_{10}	
20	Q_{20}	
30	Q_{30}	
.. .. ..		
100	Q_{100}	
110	Q_{110}	
200	Q_{200}	

TRAFFIC FLOW, CAPACITY AND PROPERTIES OF THE TRAFFIC



Rural Roadway 30th Hourly Volume (HV) = (AADT)*%12~20 30th HV $= \frac{YOGT}{6}$

Urban Roadways; 30th HV $= \frac{YOGT}{8}$

Example

- Freeway
- AADT = 45,000 veh/day
- FFS = 75 mph
- Directional Factor = 70%

Number of lanes required to provide at least LOS C using the 30th highest annual hourly volume?

Example

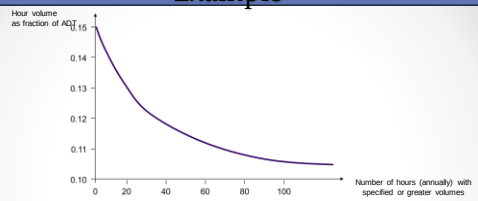


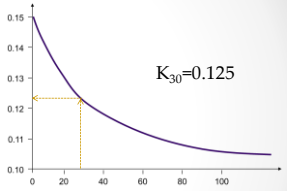
Table 6.1 LOS Criteria for Basic Freeway Segments

Criteria	LOS				
	A	B	C	D	E
	FFS = 75 mi/h				
Maximum density (pc/mi/ln)	11	18	26	35	45
Minimum speed (mi/h)	75.0	74.8	70.6	62.2	53.3
Maximum v/c	0.34	0.56	0.76	0.90	1.00
Maximum service flow rate (pc/h/ln)	820	1350	1830	2170	2400

Solution

$$DDHV = K \times D \times AADT$$

D = 0.70 (given)
AADT = 45,000 (given)



K_{30}=0.125

from graph ➔ K = 0.125

DDHV= 0.125*0.70*45000= **3937** veh/hr

Solution

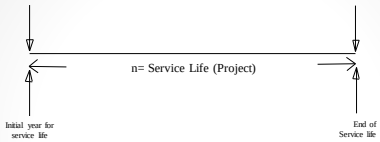
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Maximum v/c	0.34	0.56	0.76	0.90	1.00
Maximum service flow rate (pc/h/ln)	820	1350	1830	2170	2400

- N= DDHV/Q_{LOSc}

→N= 3937/ 1830= 2.15

→ **N=3**

TRAFFIC FLOW, CAPACITY AND PROPERTIES OF THE TRAFFIC

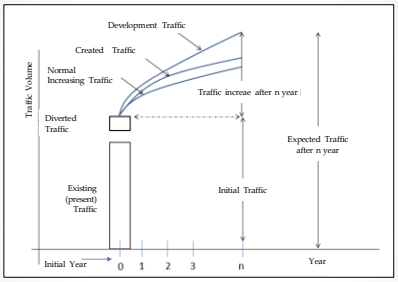


Project traffic is estimated for end of the service life.

$$(AADT)_{terminal} = (AADT)_{initial} * (1+r)^n$$

r: Trafifc increase rate (%)
n: Service life

TRAFFIC FLOW, CAPACITY AND PROPERTIES OF THE TRAFFIC



TRAFFIC FLOW, CAPACITY AND PROPERTIES OF THE TRAFFIC

The components of the expected traffic;

a) **Normal Increased (trend):** The traffic increased by the population of the zone, number of vehicles and mobility (average number of trips done each day by vehicles) is the normally increasing traffic. The determination of the normally increasing traffic is very hard because the parameters it depends on is many and different (r1).

b) **Created traffic:** The people have repressed their transportation needs due to several negative effects. As the conditions improve, people start to use the transportation system in order to satisfy their needs. Created traffic emerges within few years after the highway is put into service (r2).

c) **Development traffic:** It is already stated that a highway boosts the route that it passes from. The traffic generated by the land use changes, increament of some activities after a highway is put into the service, is named as the development traffic (r3).

d) **Diverted traffic:** The traffic diverted from other highways to the highway to be constructed is the diverted traffic. (r4).

Toplam trafik artış yüzdesi:
r = r1+r2+r3+r4